SHAOQING DAI

WORKING EXPERIENCE

Current | 2025.01

Key-Funded Postdoctoral Researcher

Geography Postdoctoral Program, School of Resource and Environmental Sciences, Wuhan University

Wuhan, China

- Co-mentor: Prof. Peng Jia

EDUCATION

2024.06 | 2020.01

Ph.D., Geoinformatics

Faculty of Geo-information Science and Earth Observation (ITC), University of Twente

Enschede, Netherlands

- Thesis: Improving obesogenic environmental assessments with advanced geospatial methods
- Supervised by Prof. Peng Jia, Prof. Alfred Stein

2019.06 | 2016.09

M.Sc., Ecology

Institute of Urban Environment, Chinese Academy of Sciences, University of Chinese Academy of Sciences

Xiamen, China

- Thesis: Creating a Carbon Source and Sink Map by Coupling an Ecological Process Model with an Emission Inventory to Study a Carbon Balance
- Supervised by Prof. Yin Ren

2016.06 | 2012.09

B.S., Geographical Science (National Talents Training Base)

School of Geographical Sciences, Fujian Normal University — Fuzhou, China

- Thesis: The Planning Location and Design of Smart Campus Based on BIM and GIS: A Case Study of Fujian Normal University
- Supervised by Prof. Huixian Jiang

RESEARCH INTERESTS

Current | 2012

Health Geography; Urban Visual Intelligence; Spatial Lifecourse Health; Spatial Statistics; GeoAI; Spatial-Temporal Big Data



View this CV online with links at *nickstrayer.me/cv/*

CONTACT

shaoqing.dai@outlook.com

GISerDaiShaoqing

Homepage

LANGUAGE SKILLS

- R
- GIS
- 3D Modeling
- Python
- Statistics
- RS & Earth Observation

*Made with the R package **pagedown**.*

The source code is available on GISerDaiShaoqing/Curriculum-Vitae-Latex.

Last updated on 2026-11-02.

DEVELOPED OPEN SOFTWARE

Current | 2017

Note of Applied Statistics with R

- http://gisersqdai.top/Note-of-Applied-Statistics-with-R-Book/

Current | 2018

D3L Tool of NASA Satellite

- The program provides a graphical interface to download satellite imagery from NASA, such as MODIS products and MERIS; for more information, see url[1]

Current | 2018

Geographical detector in R

- The geographical detector software based on R

Current | 2018

GitHub Paper Notebook

- The software for notebook of papers

Current | 2019

rgeoda

- rgeoda is an R package for spatial data analysis based on libgeoda and GeoDa (contributor).

Current | 2020

COVID-19 Dashboard shiny app

- A developed dashboard shiny app about COVID-19, url[2].

Current | 2020

Awesome COVID-19

- Resources of COVID-19 research. http://covid19.gisersqdai.top/en-us/

Current | 2023

MetS score calculator shiny app

- A shiny app to calculate MetS score. https://gisersqdai.shinyapps.io/mets_severiity_calculator/

AWARDS & HONORS

2025

Best Oral Presentation Award in 3rd Urban Agglomeration Green Development Conference

- Urban Agglomeration Green Development Special Committee, Chinese Society for Urban Studies

2025 **The National 2nd Prize in the 9th Planning Decision Support Model Design Contest [3]Supervisor**

- Beijing Chengyuan Digital Technology Co. Ltd., World Urban Planning Education Network, Beijing Planning Institute Hongdu Planning and Architectural Design Research Institute Co. Ltd.

2025 **Special Program for the Introduction of Overseas Postdoctoral Talents**

- Ministry of Education of the People's Republic of China

2022 **Outstanding contributions to COVID-19 epidemic prevention and control**

- Health commission of Tumot Left Banner, Inner Mongolia Autonomous Region

2021 **The National 2nd Prize in the 5th Planning Decision Support Model Design Contest (Chengyuan Cup)**

- Beijing Chengyuan Digital Technology Co. Ltd., World Urban Planning Education Network, Beijing Planning Institute Hongdu Planning and Architectural Design Research Institute Co. Ltd., Beijing City Lab, Baidu Map Huiyan, China Unicom Smart Footprint

2019 **China Scholarship Council (CSC)**

- China Scholarship Council

2019 **The 2nd Grade Scholarship of Yonker Environmental Protection**

- Yonker Environmental Protection Co. Ltd.

2019 **The 1st Grade Scholarship of Institute of Urban Environment, Chinese Academy of Sciences**

- Institute of Urban Environment, Chinese Academy of Sciences

2019 **The Scientific and Technological Innovation Award of Outstanding Graduate Students of the Ecological Society of China**

- Ecological Society of China

2018 **National Scholarship for M.Sctop 2-3%**

- Ministry of Education of the People's Republic of China

2018 **The National 3rd Prize in the 2nd Big Data Support Spatial Planning and Design Competition**

• Urban Big Data Special Committee, Chinese Society for Urban Studies

2018 **“Applied Energy”2018 Summer School Certification**

• The Applied Energy Innovation Institute, Department of Earth System Science, Tsinghua University, China Emission Accounts & Datasets (CEADs)

2018 **Merit Students in the academic year 2017-2018**

• University of Chinese Academy of Sciences

2018 **The 2nd Grade Scholarship of Institute of Urban Environment, Chinese Academy of Sciences**

• Institute of Urban Environment, Chinese Academy of Sciences

2018 **The 1st Scholarship of Institute of Urban Environment, Chinese Academy of Sciences**

• Institute of Urban Environment, Chinese Academy of Sciences

2018 **The National Winning Award in 2017 “Sharing Cup” College Student Science and Technology Resource Sharing Service Innovation Contest**

• National Science & Technology Infrastructure Center

2017 **“Excellent Volunteer” award of 15th User Conference of Esri China**

• Esri China Information Technology Co. Ltd.

2017 **Merit Students in the academic year 2016-2017**

• University of Chinese Academy of Sciences

2015 **The National Winning Award in 2015 Esri Cup National Collegiate Development Contest in GIS·3D Modeling and Design**

• Chinese Society of Geodesy Photogrammetry and Cartography, Esri China Information Technology Co. Ltd.

2015 **Innovation Scholarships in the academic year 2014-2015**

• Fujian Normal University

2015 **The National 3rd Prize in the Smart City Special Event Contest, the 14th Challenge Cup National College Students' Extracurricular Academic Science and Technology Works Contest**

- Chinese Communist Youth League, China Association for Science and Technology, Ministry of Education of the People's Republic of China, All-China Students' Federation

2015 **Scientific and Technological Innovation award and Nomination, the First Annual Fujian Normal University Student Selection**

- Fujian Normal University

2015 **The School Wining Award during the 9th Challenge Cup National College Students' Extracurricular Academic Science and Technology Works Contest**

- Fujian Normal University

2014 **Innovation Scholarships in the academic year 2013-2014**

- Fujian Normal University

2014 **The National 3rd Prize in 2014 MathorCup Global Collegiate Mathematical Contest in Modeling**

- Chinese Society of Optimization, Overall Planning and Economical Mathematics

2014 **"Excellent League Cadres" in the academic year 2013-2014**

- Fujian Normal University

2014 **"Excellent League Cadres" in the academic year 2013-2014**

- School of Geographical Sciences, Fujian Normal University

2013 **"Excellent League Cadres" in the academic year 2012-2013**

- School of Geographical Sciences, Fujian Normal University

ACADEMIC SERVICE & AFFILIATIONS

Current | 2026 **Editors**

npj Digital Public Health

- Editorial Board Member

Current | 2026

Editors

Journal of Subtropical Resources and Environment

• Youth Editorial Board Member

Current | 2025

Editors

BMC Public Health

• Editorial Board Member

Current | 2025

Editors

Frontiers in Public Health, Frontiers in Psychology, Frontiers in Forests and Global Change

• Associate Editor

Current | 2025

Editors

Land Special Issue, "Urban Spatial Planning for Health and Well-Being"

• Guest Editor

Current | 2024

Affiliations

AI 4D City lab, Hong Kong University of Science and Technology (Guangzhou)

• Visiting Scholar

Current | 2019

Affiliations

International Institute of Spatial Lifecourse Health (ISLE)

• Fellow

Current | 2019

Affiliations

GeoDa Center, rgeoda, libgeoda Github Organizations

• Contributor

Current | 2015

Social Media Volunteer

The Organization "Sustainable Cities & Mobility"

• Volunteer

Current | 2025

Science & Technology Service

Pang Changle Expert Grassroots Scientific Research Workstation, Mengzi Shuchan Investment and Development Co. Ltd

• Chief Researcher (PI)

Current | 2025

Conference & Society Service

The Geographical Society of China

• Member

Current | 2024

Conference & Society Service

International Society for Photogrammetry and Remote Sensing Student Consortium

• Member

Current | 2024

Workshop Lecturer

UrbanParty

• Lecturer

Current | 2019

Peer-Reviewed Journal & Conference Reviewer

Health & epidemiology: European Heart Journal, BMC Public Health, Frontiers in Public Health, HealthCare, Children, Artificial Intelligence in Emergency Medicine, Biology and Life Sciences Forum, International Journal of Environmental Research and Public Health; GIS & remote sensing: GIScience & Remote Sensing, International Journal of Geographical Information Science, International Journal of Digital Earth, Spatial Economic Analysis, Spatial Statistics, Geographies, Information Geography, ISPRS Conference, CPGIS Conference; environment & ecology: Ecological Indicators, Environmental Research, Methods in Ecology and Evolution, City and Environment Interactions, Environmental Modelling & Software, Frontiers in Environmental Science, Frontiers in Forests and Global Change, Forest Ecology & Management, Regional Environmental Change, Atmospheric Pollution Research, Environmental Modeling & Assessment, Frontiers in Earth Science; urban & built environment: Sustainable Cities and Society, Habitat International, Cities, Frontiers of Architectural Research, Journal of Asian Architecture and Building Engineering, Urban & Building Science, Discover Cities, Shanghai Urban Planning Review; energy, resources & carbon: Applied Energy, Resources, Conservation and Recycling, Carbon Balance and Management, Carbon Management; data, computing & open science: Information Processing & Management, Scientific Data, Sensors, Data in Brief, The Journal of Open Source Software, Transactions in Urban Data, Science, and Technology; multidisciplinary & others: Scientific Reports, Sci, Sustainability, Applied Economics Letters, Heliyon, Discover Global Society

• Reviewer

2026 | 2025

Editors

Frontiers in Remote Sensing Special Issue, "Machine Learning for Advanced Remote Sensing: From Theory to Applications and Societal Impact"

• Guest Editor

2026

Conference & Society Service

GISphere GISsalon No.49, GIS×Health

• Panelist

2026

Workshop Lecturer

6th International Symposium on Spatial Lifecourse Health Workshop

• Lecturer

2025

Science & Technology Service

The 138th Postdoctoral Science and Technology Service Corps, China Postdoctoral Science Foundation

• Fellow

2025

Conference & Society Service

2nd National Conference of Information Geography, "Human behaviors, health geography and spatial modeling" session

• Convener

2025

Conference & Society Service

5th International Symposium on Spatial Lifecourse Health, "Street view images, human mobility and environmental health" session

• Convener

2025

Workshop Lecturer

5th International Symposium on Spatial Lifecourse Health Workshop

• Lecturer

2024

Conference & Society Service

2nd World Congress of Geography, "Research on Global Health Wellbeing and Health Geography" session

• Convener

2024

Conference & Society Service

4th International Symposium on Spatial Lifecourse Health, "Application of System Simulation Models in Health Domain" session

• Convener

2024

Workshop Lecturer

4th International Symposium on Spatial Lifecourse Health Workshop

• Lecturer

2024

Workshop Lecturer

Heywhale Community and GISphere, 'Python x GIS' GeoAI Workshop

• Lecturer

2024

Workshop Lecturer

Training course on 'Spatial Epidemiology Analysis Methods and Applications'

• Lecturer

2023 | 2020

Social Media Volunteer

GISphere

• Editorial Board, Council Member, Editor in chief of Wechat

2020 | 2018

Science & Technology Service

Aliyunqi Community

- Expert

2019 | 2017

Conference & Society Service

International Association of Landscape Ecology of China (IALE-China)

- Member

2017 | 2013

Editors

The Geographical Tribune of Youth

- Founder, Editor in Chief, Associate Editor, Assistant Editor in Chief

2017 | 2016

Social Media Volunteer

Green Bike-Transit

- Volunteer

PUBLICATIONS-PEER-REVIEWED JOURNAL ARTICLES

2026

Co-exposure to CO2 and heatwaves and the risk for diabetes: a multi-center spatial epidemiological study in China

BMC Medicine

- Manyao Li, Lan Zhang, Yuanyuan Shi*, Shuhan Yang, Kun Qin, Zhanpeng Wang, Chen Li, Bin Yu, **Shaoqing Dai**, Xiaoqing Li, Ying Shao, Shujuan Yang, Chun Yin, Peng Jia*
-
- doi:10.1186/s12916-026-05197-w.
- SCI, IF=8.7

2026

Evaluating the Marginal Contribution of Remote Sensing for Forest Biomass Estimation When Inventory Data Exists

Forests

- Xiaoman Zheng, Ying Su, Yunxia Wang, **Shaoqing Dai**, Yufeng Chi, Guanjun Lin, Yin Ren*
- 17(9): 1094. doi:10.3390/f17091094.
- SCI, IF=3.1

2026

Toward precision urban resilience: Integrating multimodal large language models with spatial analytics for fire risk governance

Cities

- Dong Xie, Senhong Cai, Ke Wang, Ce Hou, **Shaoqing Dai**, Jingnan Huang*
- 179: 107449. doi:10.1016/j.cities.2026.107449.
- SSCI, IF=7.0

2026

High-resolution mapping of demographic compositions based on remote sensing and social media data: a multimodal, multi-output modeling approach

International Journal of Geographical Information Science

- Ge Qiu, Yuchen Li, **Shaoqing Dai**, Kun Qin, Zhanpeng Wang, Yaolin Liu, Jiping Liu*, Peng Jia*
- online first: 1-35. doi:10.1080/13658816.2026.2702485.
- SCI/SSCI, IF=5.9

2026

Editorial: Machine learning for advanced remote sensing: from theory to applications and societal impact

Frontiers in Remote Sensing

- Rui Li*, **Shaoqing Dai**, Bin Jiang, Cong Zhang, Haoyang Yang, Wufan Zhao
- 7: 1931394. doi:10.3389/frsen.2026.1931394.
- ESCI, IF=3.5

2026

Measuring causal strengths from spatial cross-sectional data with geographical cross mapping cardinality

International Journal of Geographical Information Science

- Wenbo Lyu, **Shaoqing Dai**, Yongze Song*, Wufan Zhao, Wen Yi, Yumiao Xiao, Nan Jia
- online first: 1-23. doi:10.1080/13658816.2026.2687121.
- SCI/SSCI, IF=5.9

2026

Understanding multi-perspective urban green space patterns under urban expansion: Evidence from Guangzhou

Sustainable Cities and Society

- Qiming Liu, **Shaoqing Dai**, Guosheng Yang, Cai Wu*
- 147: 107534. doi:10.1016/j.scs.2026.107534.
- SCI, IF=13.3

2026

Bridging street view coverage disparities through geographic identity preserving generation from satellite view

ISPRS Journal of Photogrammetry and Remote Sensing

- Zongrong Li, Fan Zhang, **Shaoqing Dai***, Wufan Zhao*
- 236: 622-639. doi:10.1016/j.isprsjprs.2026.03.049.
- SCI, IF=12.9

2026

Causal discovery in urban data with temporal empirical dynamic modeling: The R package tEDM

Computers, Environment and Urban Systems

- Wenbo Lyu, Yangyang Lei, Wen Yi, Yongze Song, Xiao Li, **Shaoqing Dai**, Yiming Qin, Wufan Zhao
- 127: 102435. doi:10.1016/j.compenvurbsys.2026.102435.
- SSCI, IF=8.9

2025

Effects of short-term exposure to air pollutants on real-time blood pressure: a wearable device-based study in China

Journal of Urban Health

- Chen Li, Xiu Yang, Kun Qin, Shuhan Yang, **Shaoqing Dai**, Chun Yin, Shujuan Yang, Yuanyuan Shi, Peng Jia
- 102(5): 1011-1023. doi:10.1007/s11524-025-01017-3.
- SCI, IF=3.2

2025

Assessing variations in river networks under urbanization across metropolitan plains using a multi-metric approach

Land

- Zhixin Lin, Shuang Luo, Miao Lu, **Shaoqing Dai**, Youpeng Xu
- 14(10): 1994. doi:10.3390/land14101994.
- SSCI, IF=3.5

2025

Agricultural socialized service system and its mechanism driving industrial upgrade: based on an empirical study on value chain theory

Resources & Industries

- Yuxei Lin, Feng Lu, Zhengyi Xu, Junfeng Xue, **Shaoqing Dai**, Xinying Zhang
- 7(05): 139-150. doi:10.13776/j.cnki.resourcesindustries.20250901.006.
- in Chinese

2025

High-resolution population density mapping in urban areas using a contextualized geographically weighted neural network (CGWNN) model

Applied Geography

- Ge Qiu, Yuchen Li, Kun Qin, Chen Li, Shuhan Yang, Chun Yin, Yaolin Liu*, **Shaoqing Dai***, Peng Jia*
- 182: 103708. doi:10.1016/j.apgeog.2025.103708.
- SSCI, IF=7.4

2025

Identifying landscape patterns at different scales asdriving factors for urban flooding

Ecological Indicators

- Yao Li, Frank Badu Osei, **Shaoqing Dai**, Tangao Hu, Alfred Stein.
- 176: 113614. doi:10.1016/j.ecolind.2025.113614.
- SCI, IF=8.7

2025

Built environments and obesity: A framework considering residences, commute routes, and workplaces

Cities

- Chun Yin, Zhe Zhang, **Shaoqing Dai**, Yiyi Chen
- 160: 105842. doi:10.1016/j.cities.2025.105842.
- SSCI, IF=7.0

2025

Toward 3D hedonic price model for vertically developed cities using street view images and machine learning methods

Habitat International

- Yue Ying, **Shaoqing Dai***, Mila Koeva, Monika Kuffer, Claudio Persello, Wen Zhou, Jaap Zevenbergen
- 156: 103288. doi:10.1016/j.habitatint.2025.103288.
- SSCI, IF=8.5

2025

Heterogeneous impacts of and vulnerabilities to the COVID-19 pandemic

Landscape Ecology

- Manyao Li#, **Shaoqing Dai#**, Yuanyuan Shi#, Kun Qin#, Ross Brownson, Yan Kestens, Miyang Luo, Shiyong Liu, Jing Su, Gordon G. Liu, Shujuan Yang, Peng Jia
- 40: 32. doi:10.1007/s10980-024-02039-z
- SCI, IF=4.7

2024

Long-term exposure to PM2.5 chemical constituents and diabesity: evidence from a multi-center cohort study in China

The Lancet Regional Health–Western Pacific

- Changwei Cai, Shuzhen Zhu, Mingfang Qin, Xiaoqing Li, Chuanteng Feng, Bin Yu, **Shaoqing Dai**, Ge Qiu, Yuchen Li, Tingting Ye, Wenling Zhong, Ying Shao, Lan Zhang, Peng Jia, Shujuan Yang
- 47: 101100. doi:10.1016/j.lanwpc.2024.101100
- SCI/SSCI, IF=9.4

2024

Street view imagery-based built environment auditing tools: a systematic review

International Journal of Geographical Information Science

- **Shaoqing Dai**, Yuchen Li, Alfred Stein, Shujuan Yang, Peng Jia.
- 38(6): 1136-1157. doi:10.1080/13658816.2024.2336034
- SCI/SSCI, IF=5.9

2024

Spatiotemporal patterns of air pollutants over the epidemic course: a national study in China

Remote Sensing

- Kun Qin#, Zhanpeng Wang#, **Shaoqing Dai#**, Yuchen Li, Manyao Li, Chen Li, Ge Qiu, Yuanyuan Shi, Chun Yin, Shujuan Yang, Peng Jia*
- 16(7): 1298. doi:10.3390/rs16071298.
- SCI, IF=4.3

2024

State of the art of lifecourse cohort establishment

China CDC Weekly

- **Shaoqing Dai**, Ge Qiu, Yuchen Li, Shuhan Yang, Shujuan Yang, Peng Jia
- 6(14), 300–304. doi:10.46234/ccdcw2024.058.
- SCI/SSCI, IF = 3.5

2024

Long-term dynamic monitoring and driving force analysis of eco-environmental quality in China

Remote Sensing

- Weiwei Zhang, Zixi Liu, Kun Qin, **Shaoqing Dai**, Huiyuan Lu, Miao Lu, Jianwan Ji, Zhaohui Yang, Chao Chen, Peng Jia
- 16(6): 1028 doi:10.3390/rs16061028.
- SCI, IF=4.3

2024

Mixed land use has opposite associations with subjective well-being through social capital: Spatial heterogeneity in residential and workplace neighborhoods

Travel Behaviour and Society

- Chun Yin, Jiahang Liu, **Shaoqing Dai**, Bindong Sun
- 35, 100724. doi:10.1016/j.tbs.2023.100724.
- SSCI, IF=5.8

2024

Local carbon emission zone construction in the highly urbanized regions: Application of residential and transport CO_2 emissions in Shanghai, China

Building and Environment

- Yunfeng Tian, Shudi Zuo, Jiaheng Ju, **Shaoqing Dai**, Yin Ren, Panfeng Dou.
- 247, 111007. doi:10.1016/j.buildenv.2023.111007.
- SCI, IF=8.4

2023

Assessing Spatiotemporal Bikeability Using Multi-Source Geospatial Big Data: A Case Study of Xiamen, China

International Journal of Applied Earth Observation and Geoinformation

- **Shaoqing Dai**, Wufan Zhao, Yanwen Wang, Xiao Huang, Zhidong Chen, Jinghan Lei, Alfred Stein, Peng Jia.
- 125, 103539. doi:10.1016/j.jag.2023.103539.
- SCI, IF=8.2

2023

Development and validation of an age-sex-ethnicity-specific metabolic syndrome score in the Chinese adults

Nature Communications

- Shujuan Yang, Bin Yu, Wanqi Yu, **Shaoqing Dai**, Chuanteng Feng, Ying Shao, Xing Zhao, Xiaoqing Li, Tianjing He, Peng Jia.
- 14(1), 6988. doi:10.1038/s41467-023-42423-y
- SCI, IF=18.1

2023

Environmental determinants of childhood obesity: a meta-analysis

The Lancet Global Health

- Peng Jia, Yuanyuan Shi, Qi Jiang, **Shaoqing Dai**,Bin Yu, Shuhan Yang, Ge Qiu, Shujuan Yang.
- 11, S7. doi:10.1016/S2214-109X(23)00092-X.
- SCI/SSCI, IF=22.5

2022

The importance of the functional mixed entropy for the explanation of residential and transport CO_2 emissions in the urban center of China

journal of Cleaner Production

- Shudi Zuo, **Shaoqing Dai**, Jiaheng Ju, Fanxin Meng, Yin Ren, Yunfeng Tian, Kaide Wang
- 380, 134947. doi:10.1016/j.jclepro.2022.134947.
- SCI, IF=10.7

2022

Global spreading of Omicron variant of COVID-19

Geospatial Health

- Pengxin Zhang, Shuhan Yang, **Shaoqing Dai**, Darren How Jin Aik, Shujuan Yang, Peng Jia.
- Geospatial Health. 17. doi:10.4081/gh.2022.1083.
- SCI, IF=1.5

2022

Quantifying urban mass gain and loss by a GIS-based material stocks and flows analysis. Journal of Industrial Ecology

Journal of Industrial Ecology

- Yupeng Liu, Jiajia Li, Weiqiang Chen, Lulu Song, **Shaoqing Dai**.
- 26(3), 1051-1060. doi:10.1111/jiec.13252.
- SCI, IF=5.7

2022

Identification of two-dimensional copper signatures in human blood for bladder cancer with machine learning

Chemical Science

- Weichao Wang, Xian Liu, Changwen Zhang, Fei Sheng, Shanjun Song, Penghui Li, **Shaoqing Dai**, Bin Wang, Dawei Lu, Luyao Zhang, Xuezhi Yang, Zhihong Zhang, Sijin Liu, Aiqian Zhang, Qian Liu, Guibin Jiang.
- 3(6), 1648-1656. doi:10.1039/D1SC06156A.
- SCI, IF=8.1

2021

Improving plot-level model of forest biomass: A combined approach using machine learning with spatial statistics

Forests

- **Shaoqing Dai**, Xiaoman Zheng, Lei Gao, Chengdong Xu, Shudi Zuo, Qi Chen, Xiaohua Wei, Yin Ren
- 12(12), 1663. doi:10.3390/f12121663.
- SCI, IF=3.1

2021

New approaches to anticipate the risk of reverse zoonosis

Trends in Ecology & Evolution

- Peng Jia, **Shaoqing Dai**, Tong Wu, Shujuan Yang.
- 36(7), 580-590. doi:10.1016/j.tree.2021.03.012.
- SCI, IF=17.9

2021

Refined water security assessment for sustainable water management: A case study of 15 key cities in the Yangtze River Delta, China

Journal of Environmental Management

- Panfeng Dou, Shudi Zuo, Yin Ren, Manuel J. Rodriguez, **Shaoqing Dai**.
- 290, 112588. doi:10.1016/j.jenvman.2021.112588.
- SCI, IF=9.2

2021

Pitfalls in Modeling Asymptomatic COVID-19 Infection

Frontiers in Public Health

- Shujuan Yang, **Shaoqing Dai**, Yulin Huang, Peng Jia.
- 9, 593176. doi:10.3389/fpubh.2021.593176.
- SCI/SSCI, IF=4.1

2021

High spatial resolution mapping of steel resources accumulated above ground in mainland China: Past trends and future prospects

Journal of Cleaner Production

- Lulu Song, **Shaoqing Dai**, Zhi Cao, Yupeng Liu, Weiqiang Chen.
- 297, 144609. doi:10.1016/j.jclepro.2021.126482.
- SCI, IF=10.7

2020

The direct and interactive impacts of hydrological factors on bacillary dysentery across different geographical regions in central China.

Science of the Total Environment

- Shudi Zuo, Lianping Yang, Panfeng Dou, Hung Chak Ho, **SQ. Dai**, Wenjun Ma, Yin Ren, Cunrui Huang.
- 764, 144609. doi:10.1016/j.scitotenv.2020.144609.
- SCI, IF=8.0

2020

Investigating the uncertainties propagation analysis of CO_2 emissions gridded maps at the urban scales: a case study of Jinjiang city, China.

Remote Sensing

- **Shaoqing Dai**, Yin Ren, Shudi Zuo, Chengyi Lai, Jiajia Li, Shengyu Xie, Bingchu Chen.
- 12(23), 3932. doi:10.3390/rs12233932
- SCI, IF=4.3

2020

The Evaluation of Health Effect of Short-term Exposure to PM2.5 during Spring Festival: A Case Study of 25 Cities in the Yangtze River Delta

Shanghai Urban Planning Review

- **Shaoqing Dai**, Jiajia Li, Weixu Yang, Fangyu Chen, Huixian Jiang.
- 05, 22-29. doi:10.11982/j.supr.20200504.
- in Chinese

2020

Natural environment and childhood obesity: A systematic review

Obesity Reviews

- Peng Jia, **Shaoqing Dai**, Kristen. E. Rohli, Robert V. Rohli, Yanan Ma, Chao Yu, Xiongfeng Pan, Weiqi Zhou.
- 1-9. doi:10.1111/obr.13097
- SCI, IF=7.6

2020

Response of syntrophic bacterial and methanogenic archaeal communities in paddy soil to soil type and phenological period of rice growth

Journal of Cleaner Production

- XiaoFang Pan, Hu Li, Lixin Zhao, Xiaoru Yang, Jianqiang Su, **Shaoqing Dai**, Jing Ning, Chunxing Li, Guanjing Cai, GeFu Zhu.
- 278, 123418. doi:10.1016/j.jclepro.2020.123418
- SCI, IF=10.7

2020

Green space access in the neighbourhood and childhood obesity

Obesity Reviews

- Peng Jia, Xinxi Cao, Hongxi Yang, **Shaoqing Dai**, Pan He, Ganlin Huang, Tong Wu, Yaogang Wang.
- 1-12. doi:10.1111/obr.13100
- SCI, IF=7.6

2020

Neighbourhood speed limit and childhood obesity

Obesity Reviews

- Miyang Luo, Hanqi Li, Xiongfeng Pan, Teng Fei, **Shaoqing Dai**, Ge Qiu, Yuxuan Zou, Heleen Vos, Jiayou Luo, Peng Jia.
- 1-11. doi:10.1111/obr.13052
- SCI, IF=7.6

2020

Treatment of high-ash industrial sludge for producing improved char with low heavy metal toxicity

Journal of Analytical and Applied Pyrolysis

- Shengyu Xie, Guangwei Yu, Chunxing Li, Jie Li, Gang Wang, **Shaoqing Dai**, Yin Wang.
- 150, 104866. doi:10.1016/j.jaap.2020.104866
- SCI, IF=6.7

2020

Access to bike lanes and childhood obesity: A systematic review and meta-analysis

Obesity Reviews

- Xiongfeng Pan, Li Zhao, Jiayou Luo, Yinhao Li, Lin Zhang, Tong Wu, Melody Smith, **Shaoqing Dai**, Peng Jia.
- 1-11. doi:10.1111/obr.13042
- SCI, IF=7.6

2020

A spatial database of CO_2 emissions, urban form fragmentation and city-scale effect related impact factors for the low carbon urban system in Jinjiang city, China

Data in Brief

- Shaoqing Dai, Shudi Zuo, Yin Ren.
- 29, 105274. doi:10.1016/j.dib.2020.105274
- ESCI, IF=1.9

2020

More fragmentized urban form more CO_2 emissions? A comprehensive relationship from the combination analysis across different scales

Journal of Cleaner Production

- Shudi Zuo, Shaoqing Dai, Yin Ren.
- 244, 118659. doi:10.1016/j.jclepro.2019.118659
- SCI, IF=10.7

2019

The impacts of climate and land use/land cover changes on water yield service in Ningbo region

Acta Scientiae Circumstantiae

- Panfeng Dou, Shudi Zuo, Yin Ren, Shaoqing Dai, Guoliang Yun
- 39(7), 2398-2409. doi:10.13671/j.hjkxxb.2019.0122
- CSCD, in Chinese

2019

A GIS-based high spatial resolution assessment of large-scale PV power generation potential and associated CO_2 emission reduction in China

Applied Energy

- Qing Yang, Tianyue Huang, Saige Wang, Jiashuo Li, Shaoqing Dai, Sebastian Wright, Yuxuan Wang, Huaiwu Peng.
- 247, 254-269. doi:10.1016/j.apenergy.2019.04.005
- SCI, IF=12.2

2019

PM2.5 spatiotemporal evolution and drivers in the Yangtze River Delta between 2005 and 2015

Atmosphere

- Guoliang Yun, Yuanrong He, Yuantong Jiang, Panfeng Dou, Shaoqing Dai.
- 10(2), 55. doi:10.3390/atmos10020055
- SCI, IF=2.6

2018

Determining the Mechanisms that Influence the Surface Temperature of Urban Forest Canopies by Combining Remote Sensing Methods, Ground Observations, and Spatial Statistical Models

Remote Sensing

- Shudi Zuo, **Shaoqing Dai**, Xiaodong Song, Chengdong Xu, Yilan Liao, Weiyin Chang, Qi Chen, Yaying Li, Jianfeng Tang, Wang Man, Yin Ren.
- 10(11), 1814. doi:10.3390/rs10111814
- SCI, IF=4.3

2018

Analysis of Heavy Metal Sources in the Soil of Riverbanks Across an Urbanization Gradient

International Journal of Environmental Research and Public Health

- Shudi Zuo, **Shaoqing Dai**, Yaying Li, Jianfeng Tang, Yin Ren.
- 15(10), 2175. doi:10.3390/ijerph15102175
- SCI, IF=4.6

2018

Influence of walking environment on robbery, snatch and theft crime in urban area, H city, China

Scientia Geographica Sinica

- **Shaoqing Dai**, Huixian Jiang, Jiajia Li, Xian Su, Juan Wu, Yin Ren.
- 38(8), 1235-1244. doi:10.13249/j.cnki.sgs.2018.08.005
- CSCD, in Chinese

2018

High-resolution mapping of direct CO_2 emissions and uncertainties at the urban scale

Proceedings of Spatial Accuracy 2018.

- **Shaoqing Dai**, Shudi Zuo, Yin Ren.
- 88-90.
- EI

2018

Improving the prediction accuracy of forest aboveground biomass benchmark map by integrating machine learning and spatial statistics

Proceedings of Spatial Accuracy 2018.

- **SQ. Dai**, Xiaoman Zheng, Shudi Zuo, Yin Ren.
- 80-83
- EI

2018

Individual and Interactive Influences of Anthropogenic and Ecological Factors on Forest PM2.5 Concentrations at an Urban Scale

Remote Sensing

- Guoliang Yun, Shudi Zuo, **Shaoqing Dai**, Xiaodong. Song, Chengdong Xu, Yilan Liao, Peiqiang Zhao, Weiyin Chang, Qi Chen, Yaying Li, Jianfeng Tang, Wang Man, Yin Ren.
- 10(4), 521. doi:10.3390/rs10040521
- SCI, IF=4.3

2018

Research on the effects of different policy tools on China's emissions reduction innovation

China Population, Resources and Environment

- Qin Ye, Gang Zeng, **Shaoqing Dai**, Fenglong Wang.
- 28(02), 115-122. doi:10.12062/cpre.20170915
- CSCD, in Chinese

2017

Research on the effects of command-and-control and market-oriented policy tools on China's energy conservation and emissions reduction innovation

Chinese Journal of Population Resources and Environment

- Qin Ye, **Shaoqing Dai**, Gang Zeng.
- 16(1), 1-11. doi:10.1080/10042857.2017.1418273.
- ESCI, IF=4.7

2017

Landscape-Scale Simulation Analysis of Waterlogging and Sponge City Planning for a Central Urban Area in Fuzhou City, China.

International Low Impact Development Conference China 2016: LID Applications in Sponge City Projects.

- **Shaoqing Dai**, Jiajia Li, Shudi Zuo, Yin Ren, Huixian Jiang.
- 251-260. doi:10.1061/9780784481042.028
- EI

2016

The Analysis of Urban Spatial Development Pattern in Beijing Based on the Big Data of Government

Geomatics World

- Miaoyi Li, **Shaoqing Dai**, Jingyuan Wang, Zhenjiang Shen.
- 23(3), 20-26. doi:10.3969/j.issn.1672-1586.2016.03.004
- JST

PUBLICATIONS-BOOK CHAPTER

2026

The evaluation of health effect of short-term exposure to PM2.5 during Spring Festival: a case study of 25 cities in the Yangtze River Delta

In Applications of Medical Big Data and Artificial Intelligence (in Chinese) edited by Xifeng Wu, Guihua Zhuang, Yan Jin, Chaolong Wang, Chen Mao, Beijing: People's Medical Publishing House. ISBN:9787117387644.

• Shaoqing Dai, Huixian Jiang

2025

Healthcare service allocation and planning & Emerging fields in health geography

In Health Geography (in Chinese) edited by Peng Jia, Beijing: China Higher Education Press. ISBN:978-7-04-063403-7

• Yuanyuan Shi, **Shaoqing Dai**, Peng Jia

2022

Assessing bicycle-sharing riding friendliness by fusing multi-source heterogeneous spatio-temporal big data

Planning Decision Support Model Design Compilation 2021 (in Chinese) edit by Beijing Municipal Institute of City Planning and Design, Beijing Chengyuan Digital Technology Co. Ltd., World Urban Planning Education Network, Beijing Planning Institute Hongdu Planning and Architectural Design Research Institute Co. Ltd., Beijing: China Architecture & Building Press. ISBN: 9787112270439

• **Shaoqing Dai**, Zhidong Chen, Jinghan Lei, Yanwen Wang

2019

Individual and Interactive Influences of Anthropogenic and Ecological Factors on Forest PM2.5 Concentrations at an Urban Scale

Advances in Quantitative Remote Sensing in China–In Memory of Prof. Xiaowen Li, edited by Shunlin Liang, Guangjian Yan, Jiancheng Shi, Remote Sensing, MDPI, 2019, 316-332. ISBN: 978-3-03897-276-1. https://doi.org/10.3390/books978-3-03897-277-8.

• Guoliang Yun, Shudi Zuo, **Shaoqing Dai**, Xiaodong. Song, Chengdong Xu, Yilan Liao, Peiqiang Zhao, Weiyin Chang, Qi Chen, Yaying Li, Jianfeng Tang, Wang Man, Yin Ren.

PUBLICATIONS-PATENT

2020

Improving the accuracy of forest biomass estimate using source analysis and machine learning

CN111814397A

• Yin Ren, Xiaoman Zheng, **Shaoqing Dai**, Qi Chen, Shudi Zuo

2019 **A biomass estimation method by fusing plot data and forest inventory data**

CN110162872A

• Yin Ren, **Shaoqing Dai**, Shudi Zuo, Xiaoman Zheng.

2018 **A prediction method of subtropical forest biomass**

CN108959705A

• Qi Chen, Yin Ren, Xiaoman Zheng, Shudi Zuo, **Shaoqing Dai**

PUBLICATIONS-SCIENTIFIC AND TECHNOLOGICAL ACHIEVEMENTS

2025 **'Environment + dual carbon goals' digital and intelligent platform development (Roadshow Signing Project).**

Strait (Jinjiang) Technology Innovation Center Co., Ltd., 3rd National Postdoctoral Innovation and Entrepreneurship Competition.

• **Shaoqing Dai**, Wufan Zhao, Xinying Zhang, Zhixin Lin, Yuanyuan Shi

2023 **Scaling up of carbon sequestration for Eucalyptus plantations based on ETKF-3DVAR hybrid data assimilation**

Institute of Urban Environment, Chinese Academy of Sciences and Xiamen University of Technology, 90000890-2349, Xiamen Municipal Bureau of Science and Technology

• Yin Ren, Shudi Zuo, Lizhong Hua, **Shaoqing Dai**

PUBLICATIONS-SOFTWARE COPYRIGHT

2025 **ISLE wearable device healthy data management platform. V 1.0**

• RN: 2025SR1194658

2019 **A water safety monitoring system based on environmental IOT data and InVEST model. V1.0**

• RN: 2019SR0517519

2018 **A water conservation service monitoring system based on environmental IOT data**

• RN: 2018SR745897

PUBLICATIONS-OPEN DATASET & CODE

2024

Code and data to support 'Street view imagery for built environment auditing: a systematic review' (Version 5)

Zenodo.

- **Shaoqing Dai**, Alfred Stein, Yuchen Li, Shujuan Yang, Peng Jia.
- doi:10.5281/zenodo.10724305

2020

GISerDaiShaoqing/3dmodelFJNU 1.0 (Version 1.0)

Zenodo.

- Shaoqing Dai
- doi:10.5281/zenodo.4277977

2020

GISerDaiShaoqing/Urban-Carbon-Dioxide-sources-gridded-maps-and-itsdetemination-in-Jinjiang-city 0.6 (Version 0.6)

Zenodo.

- **Shaoqing Dai**, Shudi Zuo, Yin Ren.
- doi:10.5281/zenodo.3566072

RESEARCH PROJECTS

2028 | 2026

Research on key technologies for low-altitude UAV river-lake inspection and environmental dynamic perception and modeling

Open Funding for Anhui Province Key Laboratory of Physical GeographicEnvironment (2025PGE006)

- PI

2027 | 2026

Research on urban pollen concentration mapping driven by the synergy of GeoAI and physical models

Open Fund of State Key Laboratory of Resources and Environmental Information System (2025)

- PI

2027 | 2025

The precise monitoring and analysis for residential environment risk factors and health effects among vulnerable populations

Special Program for the Introduction of Overseas Postdoctoral Talents

- PI

2026 | 2024

Early Identification and Screening of Risk Factors for Chronic Complications in Diabetes Patients by Combining Traditional Chinese and Western Medicine

Research Project of Collaborative Chronic Disease Management by Traditional Chinese and Western Medicine

- Investigator

2026 | 2023

Evaluation and prevention of health risks in the community home environment for the elderly and children, and development of appropriate products

National Key Research and Development Plan (2023YFC3604704)

• Investigator

2025 | 2022

The mechanism of the impact of urban neighborhood-level walkability on child obesity

National Natural Science Foundation of China (42271433)

• Investigator

2021 | 2020

Investigating the Changes of Residents' Daily Behaviors, Lifestyles, and Built Environment During the COVID-19 Pandemic

WeSpace Academic Support Plan in 2020: "Tsinghua-Tencent" Emerging Technologies Future Urban Space project

• PI

2020 | 2016

Research on the key technology research and integration demonstration of urban agglomeration ecological security in the Yangtze River Delta

National Key Research and Development Plan (2016YFC0502704)

• Investigator

2019 | 2018

The construction of Typhoon Disaster Assessment Index Based on GIS and Multi-source Remote Sensing Data in Fujian Province

Open Fund of the Big Data Institute of Digital Natural Disaster Monitoring in Fujian (NDMBD2018001)

• Investigator

2018 | 2015

Scaling up of carbon sequestration for Eucalyptus plantations based on ETKF-3DVAR hybrid data assimilation

National Natural Science Foundation of China (31470578)

• Investigator

2015 | 2014

The Optimization Model of High Education Staff's Refuge Space Under Earthquake Disaster

The Technological Innovation Plan of University Students, Fujian Normal University (cxxl-2014137)

• PI

PRESENTATIONS

2026 **Sensing healthy city: from multimodal data and GeoAI to causal inference**

4th China Symposium on Urban Sustainable Development, "Perception, modeling and prediction of urban complex systems" session

Beijing, China

2026 **Sensing healthy city: from multimodal geographic data and GeoAI to causal inference**

6th International Symposium on Spatial Lifecourse Health & 11th International Symposium on Remote Sensing Diagnosis of Environmental Health, Healthy city sensing and planning session

Wuhan, China

2026 **Sensing healthy city: from multimodal geographic data and GeoAI to causal inference**

The 10th Symposium on the Application of Geographic Information and Spatial Analysis Technologies in Public Health

Shanghai, China

2026 **Sensing urban obesogenic environments based on multimodal data and GeoAI**

2026 Remote Sensing Young Scientists Forum, Smart city and remote sensing large models session

Fuzhou, China

2025 **Sensing healthy city based on multimodal data and GeoAI**

Aerospace Symposium on Data Technology and Data Curation, DTDC 2025

Qingdao, China

2025 **Toward health geography: from the COVID-19 pandemic to Health China**

GISChat

Online

2025 **Sensing healthy city based on multimodal data and GeoAI**

3rd Urban Agglomeration Green Development Conference

Xiamen, China

2025 **Sensing healthy city based on street view imagery and GeoAI**

2nd National Conference of Information Geography

Chengdu, China

2025 **Application of night light remote sensing data in urban studies**

Urban Party

Online

2025 **Sensing healthy city based on street view imagery and GeoAI**

5th International Symposium on Spatial Lifecourse Health

Wuhan, China

2025 **Sensing healthy city based on street view imagery and GeoAI**

Annual conference of The Geographical Society of Fujian

Fuzhou, China

2025 **Sensing healthy city based on street view imagery and GeoAI**
3rd China Symposium on Urban Sustainable Development — Wuhan, China

2025 **The impact assessment of urban built environment on carbon-air-health nexus: a lifecourse perspective (invited))**
3rd China Symposium on Urban Sustainable Development — Wuhan, China

2024 **Development of the frequently used built environmental auditing and simulation tools in health domain**
4th International Symposium on Spatial Lifecourse Health — Wuhan, China

2024 **Development of the frequently used environmental exposome observation tools in health domain**
2024 Health Geography Youth Scholars Forum — Zhuhai, China

2024 **Bridging environment and human health: from earth observation to human-centered observation (invited))**
AI 4D City lab, Hong Kong University of Science and Technology (Guangzhou)
Guangzhou, China

2024 **Assessment of physical activity opportunities based on multi-source geospatial data**
9th Youth Geoscience Forum — Xiamen, China

2024 **The frequently used built environmental auditing tools in the health domain**
1st National Remote Sensing and Geographic conference — Nanjing, China

2024 **Health geography studies based on spatial lifecourse epidemiology framework**
Institute of Urban Environment, Chinese Academy of Sciences
Xiamen, China

2023 **Seeking your interest and going ahead will contribute to success (invited))**
Xiamen No.1 High School of Fujian — Xiamen, China

2022 **Do neighborhood boundaries matter for examining the associations between environmental factors and obesity (invited))**
The 29th International Conference on Geoinformatics in 2022 session 10-2 GIScience for Public Health
Beijing, China

2020

The Evaluation of Health Effect of Short-term Exposure to PM2.5 during Spring Festival: A Case Study of 25 Cities in the Yangtze River Delta (invited))

The Workshop of 16th Spatial Behavior and Planning & Digitalization and Fine-Grained Governance

Xiamen, China

2020

What is geography? From coronavirus disease 2019 to Bei Dou Systems

Tong An No.1 High School of Fujian

Xiamen, China

2018

The Scale Effect, Zoning Effect of Geographical Features, the Modified Areal Unit Problem and the Challenges of Spatial Statistics

11th Chinese R Language Conference

Beijing, China

2018

High-Resolution Mapping of Direct CO_2 Emissions and Uncertainties at the Urban Scale

13th International Symposium on Spatial Accuracy Assessment in Natural Resources and Environmental Sciences (Spatial Accuracy 2018)

Beijing, China

2017

The Environmental Effect of Urban Form on PM2.5: A Case Study of Beijing-Tianjin-Hebei Urban Agglomeration

9th Chinese Landscape Ecology workshop

Guangzhou, China

2017

The Assessment Platform of Urban Forest Ecosystem Service Based on InVEST model and IOT (invited),

2nd urban environment long-term monitor and sustainable development workshop

Shanghai, China

2016

The Criminal Geographical Analysis about Walking Environment of Urban

The Workshop of 12th Spatial Behavior and Planning & Spatial-Temporal Behavior and Social Planning Research

Beijing, China

2016

Transportation Planning of Smart City on the Basis of Data Augmentation Design Under the Perspective of Humanism: A Case Study of Fuzhou's Cangshan District

10th annual conference of International Association of China Planning

Beijing, China

TEACHING EXPERIENCES

Current | 2025

Empowering GeoAI and Health Geography with R Programming

personalized cultivation week program course, School of Geographical Sciences, School of Carbon Neutrality Future Technology, Fujian Normal University

• 2 class hours

Current | 2024

Applications of Spatial Technology and AI in Health Geography

School of Resource and Environmental Science, Wuhan University

• 4 class hours

Current | 2024

Applications of Remote Sensing technology in Public Health and Diseases Prevention

China CDC National Institute of Parasitic Diseases

• 1 class hour

Current | 2024

Applications of Spatial technology, Complex Science, and AI in Health Geography

School of Resource and Environmental Science, Wuhan University

• 9 class hours

SELECTED POSTERS

2017

Quantifying the linear and nonlinear relations between the urban form fragmentation and the carbon emission distribution

2017 AGU Fall Meeting — New Orleans, USA

2017

The relationship between the fragmented urban form landscape and the carbon emission distribution at different resolutions

9th Lecture of Modern Ecology — Shanghai, China

2016

The Simulation Analysis of Waterlogging and the Sponge City Planning Control of Central Urban Area in Fuzhou City

2016 International Low Impact Development Conference — Beijing, China